

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 2: OBJECT LABELING AND COUNTING

SLO 2: Count the number of distinct objects in binary images and evaluate the accuracy of labeling techniques.

LABELING ACCURACY SOLUTIONS

Given: Actual number of objects = 4 | Algorithm detected objects = 3

1. Is the labeling result accurate or inaccurate?

Inaccurate (Accuracy = $3 / 4 = 75\%$, showing an undercount of 1 object).

2. Explain the possible reason for the error:

Two adjacent or touching objects were likely treated as a single connected region due to inappropriate connectivity rules (e.g., using 8-connectivity instead of 4-connectivity), insufficient erosion before component labeling, or thresholding noise bridging two distinct objects together.

MATCHING SOLUTIONS

Term	Matched Description
Object Counting	B. Number of separate objects in the image
Labeling Accuracy	C. Measure of correctness of labeling
Connected Component	A. Assigning labels to connected regions
Morphological Processing	D. Helps separate or refine objects

REFLECTION QUESTION SOLUTION

Why is accurate object counting important in real-world applications such as traffic monitoring, medical imaging, or industrial inspection?

Accurate object counting is vital because miscounts directly impact decision-making and critical safety outcomes:

- **Traffic Monitoring:** Ensures accurate vehicle flow estimation, congestion management, and automated tolling.
- **Medical Imaging:** Crucial for reliable cell counting (e.g., blood cell analysis or tumor cell profiling) to diagnose diseases correctly.
- **Industrial Inspection:** Guarantees strict quality control, precise inventory management, and automated packaging without missing components.